Lesson plan · for the teacher

Charts and summaries that tell the truth

Data engineering · Level 1 · Lesson 3

Time
40 min

Materials
Board and chalk, or chart paper and markers; Rulers; The class survey from lesson 2, or a week of rain readings

Learning goals

- Choose a chart for the question.
- Work out the mean and the median, and see what one wrong value does.
- Draw a fair chart: bars from zero, a title and labels.

Activity: Our class chart

1. Warm-up (5 min): Draw two bar charts of the same numbers (52 and 58), one with the axis starting at 50 and one starting at zero. Ask: which looks like a bigger change? Which one is fair?
2. Teach (8 min): With five marks (11, 12, 13, 14, 15), find the mean and the median together. Then change 15 to 150 and work them out again: the mean jumps to 40, the median stays at 13.
3. Draw (15 min): Groups draw a bar chart of the class survey on chart paper: a title, labelled axes, the unit and bars starting at zero. One group draws it on the board.
4. Summarise (7 min): Each group works out the mean and the median of the survey numbers, and circles any value far from the rest, to check it.
5. Wrap-up (5 min): Ask which summary describes the class better, and why. Then give the exit ticket.

Prepare the 11, 12, 13, 14, 15 example on the board before the lesson. If your survey has a very large value, such as a pupil who travels for two hours, keep it: it's a good example of the median holding steady.

Discussion

- Why does one wrong value move the mean so much?
- When would you show a line chart instead of bars?
- What should every chart tell you, besides the bars?

Exit ticket

- Write five numbers whose mean and median are different, and say which one is fairer.

Try it in class tomorrow
Turn your class survey into a bar chart on the board, with the axis starting at zero. Work out the mean and the median with your pupils, then ask: which one tells our story better, and why?

Worksheet · for pupils

Charts and summaries that tell the truth

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Name _____ Class _____ Date _____

1 **To show a school's attendance week by week across a term, which chart fits?**

◯ A line chart ◯ A pie chart ◯ A bar for each pupil

2 **Match each question to the chart that fits it.**

Which of four schools had the most rain on Monday?	• • • •	A table
How did attendance change week by week this term?	• • • •	A bar chart
Exactly what did each pupil score?	• • • •	A line chart

3 **One wrong value moves the median much more than the mean.**

◯ True ◯ False

4 **Complete the sentences.**

The _____ adds every value and divides by how many there are. The _____ is the middle value once they are sorted. Bars should start at _____ .

Word bank: median · zero · total · mean · range

5 **Which of these can mislead a reader about the data? Choose all that apply.**

☐ Only the average, with no chart ☐ A bar axis that starts at 50 ☐ Bars of equal width

☐ A title, and the unit on the axis ☐ Weeks left out without saying so

Answers · for the teacher

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1 **A line chart**

The question is about change over time, so a line chart fits: one point per week, joined up, shows rises and dips at a glance.

2 **Which of four schools had the most rain on Monday? — A bar chart**
How did attendance change week by week this term? — A line chart
Exactly what did each pupil score? — A table

Bars compare separate groups, a line follows one thing over time, and a table gives exact values when people need them.

3 **False**

It's the other way round. The mean adds every value, so one wild value drags it; the median is the middle value, so it hardly moves.

4 **The mean adds every value and divides by how many there are. The median is the middle value once they are sorted. Bars should start at zero.**

The mean shares the total out equally; the median is the middle of the sorted values; and bars start at zero so their heights show the real sizes.

5 **A bar axis that starts at 50; Weeks left out without saying so; Only the average, with no chart**

A cut axis exaggerates differences, missing weeks hide problems, and an average on its own hides glitches.

For you to think about

Think of a chart or an average you've seen at school or in the news. Did it tell the whole truth? What would you ask to see?